

SHIRANJEEVI V

Senior SDET Engineer | AI-Assisted Test Automation Architect

 shiranjeevi007@gmail.com |  +91-6383778113 |  LinkedIn: [Shiranjeevi V](#)

PROFESSIONAL SUMMARY

Senior SDET Engineer with 4.5+ years architecting enterprise-scale test automation frameworks across mobile (iOS/Android), web, and API ecosystems. AI-driven automation innovator leveraging Amazon Q, GitHub Copilot, and self-healing test scripts to reduce maintenance overhead by 45%. Expert in intelligent CI/CD quality gates, integrating Jenkins, qTest, and JIRA for seamless defect tracking and automated failure analysis. Proficient in Java, Python, Node.js with proven expertise in backend validation, microservices testing, and data-driven test automation. Recognized for designing scalable frameworks supporting multilingual applications and complex regression scenarios.

CORE COMPETENCIES

Test Automation & QA	Technologies & Tools
Selenium WebDriver • Appium (iOS/Android) • Playwright • Mocha Framework • Cucumber BDD • API Automation (REST/JSON) • Database Testing (SQL/MySQL) • Backend Validation • Microservices Testing • Self-Healing Scripts • Cross-Browser Testing • Data-Driven Testing • Mobile Automation (Native, Hybrid, Web)	Java Spring Boot • Python (Flask) • Node.js • JavaScript • Amazon Q (Code Gen) • GitHub Copilot • KIRO/KODA AI Accelerators • Jenkins CI/CD • Git/GitHub • qTest • JIRA • Confluence • Docker • MySQL • PostgreSQL • Agile/Scrum • Linux • DSA (Data Structures & Algorithms)

PROFESSIONAL EXPERIENCE

Custom Software Engineering Senior Analyst(Senior Automation QA Architect) | Accenture, Bengaluru

January 2022 – Present (4.5+ Years | SDET Leadership Role)

- Architected and deployed enterprise-grade cross-platform automation frameworks (Selenium, Appium, Playwright) supporting mobile (iOS/Android), web, and REST API ecosystems with 85% regression coverage across 200+ test scenarios
- Pioneered AI-assisted testing implementation: Integrated Amazon Q and GitHub Copilot for intelligent test script generation, reducing test development time by 50% and manual maintenance overhead by 45%
- Designed self-healing test automation using KIRO/KODA accelerators: Automated failure root cause analysis, reducing defect triage time from 2 hours to 15 minutes per release cycle
- Implemented intelligent CI/CD quality gates: Unified Jenkins, qTest, and JIRA integration enabling automated defect logging with rich metadata (execution logs, stack traces, screenshots), improving defect resolution SLA by 40%
- Engineered multilingual test automation framework supporting 8+ languages with runtime language selection for global mobile applications; BDD implementation using Cucumber reduced regression execution time by 35%
- Developed backend validation and microservices testing utilities: Created mock services and data verification metrics for isolated testing, enabling parallel development and reducing critical defects by 30%
- API automation expertise: Built comprehensive REST API test suites with request/response validation against MySQL databases, achieving 95% API coverage and detecting 50+ data integrity issues pre-production
- Team collaboration & mentorship: Partnered with development teams to debug critical regressions, mentored 3+ junior QA engineers on SDET practices and framework design patterns

PROJECT HIGHLIGHTS & TECHNICAL EXPERTISE

1. Mobile Automation – Global Multilingual iOS & Android Applications

Tech Stack: Appium • Cucumber BDD • KIRO/KODA • qTest • Jenkins

- Architected multilingual automation framework supporting 8 languages with dynamic runtime language selection
- Implemented BDD-driven test development using Cucumber, improving test readability and stakeholder collaboration
- Achieved 85% functional coverage across critical user journeys; reduced regression execution time by 35%

2. Web Automation with AI-Assisted Testing & Intelligent Failure Analysis

Tech Stack: Selenium WebDriver • Amazon Q • GitHub Copilot • KODA • qTest • JIRA

- Deployed AI-assisted test generation reducing script development time by 50%
- Implemented intelligent failure analysis: Automated root cause detection, reduced triage time by 87.5% (2hrs → 15min)

- Stabilized regression suites during critical release phases; integrated with Jenkins for continuous quality assurance

3. API Automation & Backend Validation with Database Cross-Validation

Tech Stack: Mocha • Node.js • REST APIs • MySQL • Python

- Designed comprehensive REST API test suites with request/response validation and database assertions
- Built backend validation utilities ensuring data integrity across microservices; detected 50+ production-critical issues pre-release
- Achieved 95% API coverage with automated CI/CD pipeline integration for continuous build verification

4. Internal ML-Driven Test Automation Dashboard & Reporting Automation

Tech Stack: Playwright • Python (Flask) • Scikit-learn • Automation

- Developed Flask-based automation dashboard using Playwright for end-to-end test reporting automation
- Implemented ML algorithms leveraging historical test failure data to optimize test case prioritization
- Automated stakeholder email reports reducing manual reporting overhead by 60%

KEY METRICS & IMPACT

- ✓ 85% Test Code Coverage: Comprehensive automation across web, mobile, and API with 200+ test scenarios
- ✓ 50% Test Development Time Reduction: AI-assisted script generation using Amazon Q and GitHub Copilot
- ✓ 45% Maintenance Overhead Reduction: Self-healing test automation and intelligent failure analysis implementation
- ✓ 87.5% Triage Time Reduction: KODA-powered failure root cause analysis (2 hours → 15 minutes)
- ✓ 40% Defect Resolution SLA Improvement: Intelligent CI/CD gates with unified Jenkins-qTest-JIRA integration
- ✓ 35% Regression Execution Time Reduction: BDD-driven multilingual framework optimization
- ✓ 30% Critical Defect Reduction: Backend validation and microservices testing utilities
- ✓ 95% API Coverage: Comprehensive REST API test automation with database cross-validation

TECHNICAL EXPERTISE MATRIX

Test Automation Frameworks: Selenium WebDriver (Expert) • Appium iOS & Android (Expert) • Playwright (Advanced) • Mocha (Advanced) • Cucumber BDD (Expert)

AI & Intelligent QA: Amazon Q Code Generation • GitHub Copilot • KIRO/KODA AI Accelerators • Self-Healing Test Scripts • AI Failure Analysis • ML Test Prioritization

Programming Languages: Java (Expert - Spring Boot, DSA) • Python (Advanced - Flask, Scikit-learn) • JavaScript/Node.js (Advanced)

API & Backend Testing: REST API Automation • Request/Response Validation • Database Testing (MySQL, PostgreSQL) • Microservices Testing • Mock Services • Data Integrity Validation

CI/CD & Infrastructure: Jenkins (Expert - Pipeline Configuration, Job Automation) • Git/GitHub • qTest (End-to-End Test Management) • JIRA (Defect Tracking Integration) • Docker • Linux

Specializations: Cross-Platform Automation • Multilingual Test Frameworks • BDD Implementation • Backend Validation • Performance Testing • Security Testing

CERTIFICATIONS, AWARDS & RECOGNITION

Databricks – Generative AI Engineer Associate (Certified) • IBM – Data Science Professional Certificate • NPTEL – Java Programming Certification • Runner-up Smart India Hackathon 2020 (MHRD, AICTE, Persistent Systems) • LeetCode: 400+ DSA Problems Solved (Top Performer) • HackerRank: Gold Badge (Java), Silver Badge (Problem Solving)

EDUCATION

Bachelor of Engineering (B.E.) in Computer Science & Engineering • Sona College of Technology (2017–2021)

Available for Immediate Interview | Open to Remote & Relocation Opportunities